

## JUGAAD Robotics Club - Public Knowledge Base & Activity Overview

### 1. ABOUT JUGAAD ROBOTICS CLUB

JUGAAD Robotics Club at UIET focuses on robotics, embedded systems, IoT, automation, and hands-on technical learning. The club organizes workshops, competitions, technical sessions, and project showcases to encourage innovation and practical engineering skills among students.

### 2. RECRUITMENT & MEMBERSHIP

- Annual recruitment drives occur at the beginning of the academic semester (September/October).
- Open to all undergraduate students interested in robotics, software development, hardware prototyping, and electronics.
- Selection process involves a basic logic test and a practical hands-on mini-project challenge.

### 3. PUBLIC EVENTS AND WORKSHOPS

- Linux Unleashed (30 September 2024): Beginner-friendly workshop introducing Linux OS, Ubuntu setup, terminal commands, and open-source computing.
- Tinker@JUGAAD 2025 - Semiconductor Workshop (28 February 2025): Guest lecture by Prof. Amanpreet Kaur (Oakland University, USA) on modern semiconductor design and chip manufacturing trends.
- Moonshot - Chandrayaan-3 Landing Celebration (23 August 2023): Live broadcast and technical discussion celebrating ISRO's lunar landing.
- JUGAAD Events at Goonj 2024 & Utsav 2024: Interactive activities including Patience Tester, Guess the Component, Circuit Tricks, and Robo-Soccer.

### 4. COMPETITIONS & AWARDS

- Line Follower Robot (LFR) Competition at Chandigarh University: Autonomous robot showcase featuring sensor navigation.
- Techzibition @ PEC (21 January 2024): Punjab Zonals for Cognizance 2024 IIT Roorkee. Selected for zonal representation.
- Cognizance 2024 at IIT Roorkee (14-17 March 2024): National tech competition featuring Nano Navigator and Faraday Station.

### 5. FEATURED PROJECTS (PUBLIC OVERVIEW)

- Nano Navigator: Autonomous micromouse maze-solving robot.
- Faraday Station: Mobile contactless wireless EV charging concept.
- Tesla Coil Project: High-frequency wireless power transfer demonstration.
- E-Conveyor: Industrial automated conveyor belt system.
- Line Follower Robot (LFR): Sensor-guided autonomous path tracking.
- RC Car: RF remote-controlled mobile platform.